

HACKATHON SHOWCASE · 2026

ClaimBack

Never pay for stock you didn't receive.

Never forget money your supplier promised to return.

01 · RECONSTRUCT TRUTH

The 3 Truths Engine

AI matches what the supplier promised, what they billed on paper, and what actually arrived in usable condition.

Voice · Paper OCR · WhatsApp · Damage

02 · DETERMINISTIC MONEY

Zero Paise Drift

AI understands messy language; strict deterministic software computes all money in integer paise. No hallucinations.

Short delivery · Rates · Schemes · Damage

03 · PERSISTENT MEMORY

Closed-Loop Recovery

"Next invoice mein adjust kar denge" creates an open obligation. ClaimBack verifies future credit notes before closing.

Hero differentiator · Ledger across time

THE PROBLEM

Small Merchants Bleed 3–8% Margin at Receiving

Retailers make sales, yet lose their profit before goods hit shelves because procurement is handled on paper, voice & memory.

LEAKAGE #1

Short Delivery

Invoiced for 50 boxes, but delivery van only unloads 48. Driver rushes merchant to sign challan.

Result: Paying for phantom stock that can never be sold.

Loss: ₹880 - ₹2,500 / order

LEAKAGE #2

Rate Mismatch

Salesman agrees on WhatsApp to ₹428/unit, but billing desk issues invoice at printed ₹441.

Result: Silent margin erosion; merchant loses 3% markup immediately.

Loss: ₹260 - ₹1,200 / order

LEAKAGE #3

Missing Schemes

Trade deal was 10+1 free units on 50 cartons (= 5 free). Supplier only shipped 3 free units.

Result: Free entitlement never enters inventory.

Loss: ₹856 - ₹3,000 / order

LEAKAGE #4 (HERO)

Forgotten Credits

Supplier says: "Bhaiya next invoice mein adjust kar denge." A week later, new invoice arrives with zero credit.

Result: 70%+ of promised supplier adjustments are forgotten.

Loss: ₹1.8L - ₹4.5L / year

BOTTOM LINE: For a pharmacy or FMCG store doing ₹15L/month, receiving leakage represents ₹40,000 - ₹90,000 of pure lost profit every single month.

CORE INNOVATION

The "Three Truths" Framework

ClaimBack reconstructs commercial reality by triangulating three independent sources of evidence.

TRUTH 01

What Was Promised

Supplier WhatsApp messages, purchase orders, verbal rates, special manufacturer schemes (e.g. 10+1 free, 5% cash discount).

AI Extraction Role:

- Extracts agreed rate, quantities & schemes
- Flags uncertain rates for confirmation
- Links exact chat excerpt as evidence locator

TRUTH 02

What Was Billed

Paper invoice, GST tax bill, supplier challan, or PDF sent by distributor delivery driver.

AI Extraction Role:

- Multimodal OCR extracts billed lines & totals
- Understands raw product names & pack sizes
- Captures printed line discounts & invoice #

TRUTH 03

What Physically Arrived

Merchant count at the door: paid cartons, free scheme cartons, and damaged/broken units.

AI Voice/Text Role:

- Hinglish voice note: "48 peti aaye, 2 damaged"
- Auto-maps spoken terms to invoice lines
- Merchant confirms counts in 1 tap

DETERMINISTIC RECONCILIATION: Once the 3 truths are confirmed, ClaimBack calculates discrepancies mathematically. If all 3 match perfectly, it reports: **"Delivery looks correct. No claim required."**

HERO DIFFERENTIATOR

Persistent Financial Memory Across Time

A supplier promise is NOT recovery. ClaimBack remembers until verified credit arrives.

DAY 01

Discrepancy Claimed

Merchant receives delivery with shortage & rate mismatch. Potential recovery: ₹1,584.

ClaimBack drafts claim with linked evidence. Merchant taps "Approve & Send".

₹1,584

Potential Recovery

DAY 02

Supplier Commitment

Supplier replies: "Shortage approved. Next invoice mein adjust kar denge."

Generic Bot: Closes ticket.

ClaimBack: Case stays OPEN!

Registers recovery obligation: ₹1,584 outstanding on ledger.

DAY 05

Audit Next Invoice

A new invoice arrives 3 days later. ClaimBack asks: "Was promised ₹1,584 credit included?"

- Missing credit? Case stays open
- Partial credit (₹1,000)? ₹584 stays open
- Full credit (₹1,584)? Moves to verification

DAY 05 (VERIFIED)

Atomic Resolution

Credit note matches open obligation. Stored procedure updates recovery ledger.

₹1,584

VERIFIED RECOVERED

Case status: RESOLVED
Dashboard margin updated.

SYSTEM ARCHITECTURE

AI Where Messy, Deterministic Software Where Critical

AI handles unstructured human reality; deterministic code protects the merchant's money.

PROBABILISTIC · AI BOUNDARY

Natural Language & Extraction

Used strictly where merchant inputs are unstructured:

- Multimodal Invoice Extraction: Reads paper receipts, blurry photos & PDFs
- Agreement Understanding: Extracts promises from WhatsApp & SMS
- Voice Parsing: Transcribes Hinglish/local speech ("do carton kam aaye")
- SKU Semantic Matching: Maps 'MAGGI 70GX96' to 'maggi small wala'
- Supplier Reply Intent: Classifies acceptances, promises & disputes

Safety Rule: Never allowed to calculate final money.

DETERMINISTIC · SOFTWARE BOUNDARY

Financial Math & State Guards

Enforces 100% financial correctness and governance:

- Integer Paise Arithmetic: Zero floating point error ($₹428.50 = 42850$ paise)
- Scheme Math Engine: Exact math on buy-X-get-Y free entitlements
- 13-State Transition Engine: States enforced by code, not LLM prompts
- Merchant Approval Guard: `sendSupplierMessage` checks `merchant_approved_at`
- Atomic Database Ledger: PostgreSQL RPC verifies credits & obligations

Safety Rule: Every claim is mathematically verified.

END-TO-END WORKFLOW

The 6-Step Margin Recovery Loop

From unloading boxes at the storefront to recovering verified cash or replacement stock.

STEP 01 · CAPTURE

Receive Stock

Merchant snaps invoice, pastes WhatsApp quote, and speaks received counts via voice note.

STEP 02 · MATCH

Triangulate Truths

AI matches invoice line items to promised scheme terms and physical counts. Ambiguities flagged for review.

STEP 03 · RECONCILE

Calculate Leakage

Deterministic engine calculates exact shortage, rate, scheme and damage variances in integer paise.

STEP 04 · APPROVE

Merchant Review

Claim packet is auto-drafted. Merchant reviews breakdown and approves sending with 1 click.

STEP 05 · DISPATCH

Supplier Claim Sent

ClaimBack sends structured claim with invoice & photo evidence. Parses incoming supplier replies.

STEP 06 · VERIFY

Closed-Loop Recovery

Next invoice / credit note is audited. Case resolves ONLY after verified cash or replacement is delivered.

EVIDENCE GROUNDING

Every Rupee Claimed is Sourced from Concrete Evidence

No black box hallucinated numbers. Discrepancy cards show exact formulas and linked artifacts.

RATE MISMATCH

₹260

Dolo 650mg Strips (20 pk)

Promised: ₹428.00 · WhatsApp quote

Billed: ₹441.00 · Invoice #INV-3812

Deterministic Formula: $(₹441 - ₹428) \times 20 \text{ usable units} = ₹13 \times 20 = ₹260$

Evidence: Verified WhatsApp Screenshot

SHORT DELIVERY

₹882

Augmentin 625 (2 Boxes)

Billed: 50 boxes · Invoice #INV-3812

Received: 48 boxes · Voice confirmation

Deterministic Formula: $(50 \text{ billed} - 48 \text{ arrived}) \times ₹441 = 2 \text{ missing} \times ₹441 = ₹882$

Evidence: Delivery Challan & Note

MISSING SCHEME UNITS

₹856

10+1 Deal on 50 Boxes

Entitled Free: 5 units $(50 \div 10)$

Delivered Free: 3 units · 2 Missing

Deterministic Formula: $2 \text{ missing free units} \times ₹428 \text{ agreed} = ₹856 \text{ unfulfilled scheme value}$

Total Case Recovery: ₹1,998

TRUST & RELIABILITY

Autonomous, But Safely Guardrailed

Small merchants cannot afford false claims that damage supplier relationships. Four built-in safeguards ensure trust.

SAFEGUARD 01

Clean Delivery = Success Outcome

If promised, billed, and received all match, ClaimBack explicitly displays: "Delivery looks correct. No claim required." Zero manufactured claims to force demo usage.

SAFEGUARD 02

Merchant Approval Guard

Autonomous drafting is safe; sending external supplier claims is high-stakes. `sendSupplierMessage` strictly requires verified `merchant_approved_at` timestamp.

SAFEGUARD 03

Uncertainty Surfaces for Review

If an invoice line is blurry or a product name is ambiguous, ClaimBack does not guess silently. It halts reconciliation and prompts merchant confirmation.

SAFEGUARD 04

Immutable Audit Trail

Every tool call logs an event into `case_events`. The UI displays an operational timeline (e.g. 'Invoice understood', 'Claim sent') without exposing raw LLM thought chains.

EVALUATION & BENCHMARKS

Measured Results Across 25 Realistic Scenarios

We don't invent benchmark numbers. Every figure below is measured by our automated test & evaluation suite.

158/158

Test Suite Passing

33 test files covering unit, integration, RLS policies, state machines & tool loops.

100% Pass Rate

25/25

Exact Money Calculations

Zero parse mathematical error across complex multi-discrepancy scenarios.

Zero Floating Drift

0/3

Clean Delivery False Claims

Clean orders produced exact "No claim required" outcomes. Zero false positives.

Perfect Clean Detection

33/33

Source-Grounded Records

Every single flagged variance carries verifiable evidence locators and excerpts.

100% Traceability

EVALUATION RUNNER INCLUDED IN REPO: Run `pnpm eval` anytime to verify 25 scenarios, SKU classification, ambiguous match detections (3/3), and multi-step recovery state progression (13/13).

DATA NETWORK MOAT

Supplier Intelligence: Turning Receiving into Leverage

Every delivery audited builds an objective supplier scorecard. Small merchants gain enterprise bargaining power.

METRIC 01

Delivery Accuracy Rate

Measures the percentage of deliveries from each distributor that arrived 100% clean without missing or broken units.

Merchant Power:

Know instantly which supplier is dependable vs which distributor consistently shorts order quantities.

METRIC 02

Recovery Velocity

Tracks average days to settle claims and fulfill promised adjustments.

Merchant Power:

Identify distributors who take 40+ days to credit shortages, versus responsive vendors who settle in 3 days.

METRIC 03

Recovery Realization %

Compares total ₹ claimed against actual verified ₹ recovered in cash or credit notes.

Merchant Power:

Brings hard data to quarterly distributor contract renewals and trade scheme negotiations.

THE COMPOUNDING MOAT: A single merchant has little leverage against regional distributors. But ClaimBack aggregates supplier reliability across thousands of merchants, transforming regional procurement transparency.

ECOSYSTEM & EXPANSION

Beyond Payments: Protecting the Entire Margin

Fintechs solved merchant payment collection. The next frontier is procurement margin protection.

TARGET PERSONA

High-Frequency Merchants

- Pharmacies: 5,000+ SKUs, daily distributor deliveries, complex schemes (10+1, 20+3)
- FMCG Grocers: High turnover, tight margins, frequent rate fluctuations
- Electronics & Hardware: Bulky items, transit damages, credit promises

TAM: 15 Million+ independent merchants in India alone.

DISTRIBUTION THESIS

Merchant Ecosystem Fit

- Paytm for Business surface: Camera & invoice workflow already in merchant hands.
- Soundbox audio notifications: "Sharma Ji, supplier Mehta owes ₹1,584 credit on today's delivery."
- Natural upsell: From collecting customer payments to protecting supplier disbursements.

Standalone MVP today; massive distribution leverage tomorrow.

FINANCIAL IMPACT

Direct Margin Expansion

- Protects 3% to 6% of procurement spend from unrecovered leakage.
- Directly increases merchant bottom-line net profit by 20% to 35%.
- Frees locked working capital without requiring business debt or loans.

ROI: 10x - 20x return on software investment within month 1.

SUMMARY

ClaimBack: The Bottom Line

Autonomous Margin Protection that doesn't stop until the money is verified in the bank.

01 · THREE TRUTHS

Full Ground Truth

Bridges messy paper invoices, WhatsApp messages and merchant voice receiving notes into verifiable facts.

Zero hallucinated claims; clean deliveries produce 'No claim required'.

02 · PERSISTENT MEMORY

Closed-Loop Recovery

Never forgets a supplier promise across days or weeks. Checks subsequent invoices and closes only when verified.

The ultimate hero differentiator that generic LLM chat products cannot offer.

03 · PRODUCTION READY

Tested & Deployed

158 passing tests, 25/25 verified benchmark scenarios, Cloudflare Workers deployment, full Supabase backend.

Built with Next.js App Router, Vercel AI SDK, and integer parse arithmetic.

EXPERIENCE THE LIVE DEMO: <https://claimback.claimback-app.workers.dev>

Upload an invoice, speak receiving counts, watch discrepancies calculate, and verify recovery in real-time.